

Assignment #03

Course: CHE 433: PROCESS SIMULATION WITH ASPEN Plus
Term: Fall 2025
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Due Date: Friday, 21st TA's Mailbox by 3:00PM

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.

Include this cover page with your name(s) in the report. Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.

Also place a hard copy of your report as a PDF file by the Due Date and Time in the TA's mailbox



- Assignments that are not typed will not be accepted.

1-PFD

This report presents an Aspen Plus simulation of a biomass pyrolysis process designed to produce liquid bio-oil. Pyrolysis is a process where biomass is heated without oxygen, causing it to break down into useful fuels such as bio and char. The simulation uses two connected fluidized bed reactors. The first reactor heats the biomass and converts it into vapors, gases, and char. The second reactor combusts the leftover char to supply heat to the system. After decomposition and combustion, solids and gases are separated using cyclone units allowing the sand and char to be removed from the product vapors.

All operating conditions and product yield data used in the simulation were taken directly from the assignment specifications to ensure consistency with the defined process. The objective of the simulation is to calculate the total energy demand of the system, evaluate steam generation at both high- and low-pressure level, and determine how the biomass is divided among the bio-oil, the combustion gases released from char burning, and the waste stream.

Component Selection in Aspen Plus

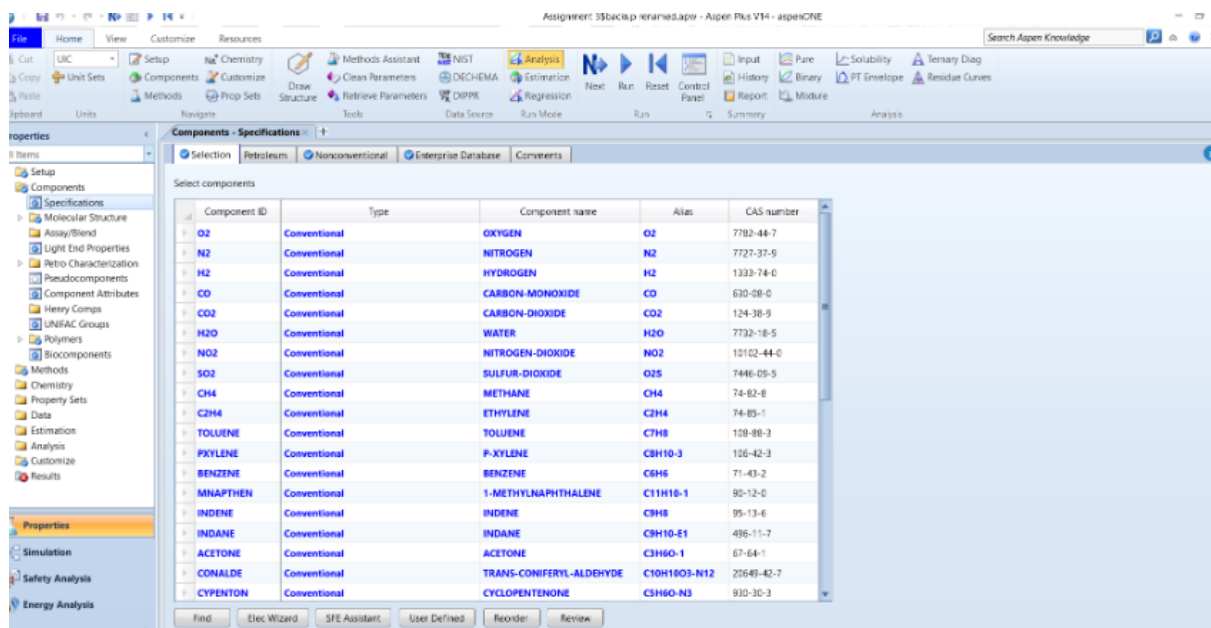


Figure 1 Components

In figure 1, the component selection window in Aspen Plus was used to put the components. The Components were taken directly from the assignment specifications and include all conventional gaseous, liquid, and combustion products. By defining these species, Aspen can apply the correct physical and thermodynamic properties.

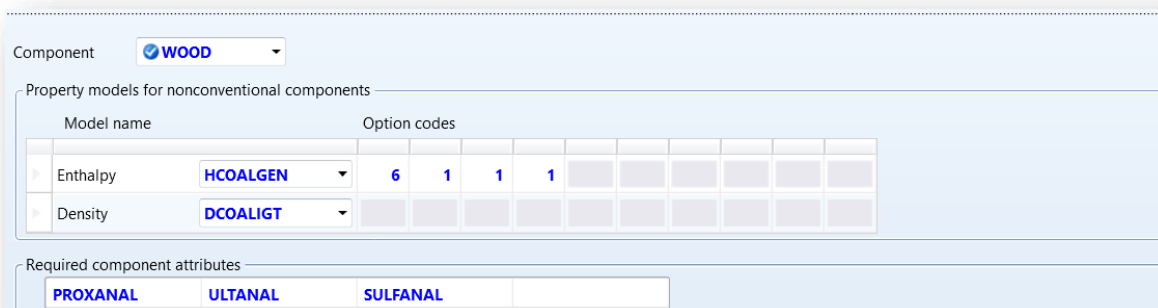


Figure 2 Enthalpy Calculation

Due to the presence of vapors, liquids, and solid fuels in the pyrolysis system, different property models are needed in Aspen Plus to correctly represent their thermodynamic behavior. Figure 2 shows the setup that is used for nonconventional components such as biomass, char, and ash. The HCOALGEN is applied to estimate the enthalpy of solid fuels based on their composition. DCOALIGT is used to estimate the density of biomass and char based on the PROXANAL, ULTANAL, and SULFANAL fuel analyses entered in Aspen. Using these nonconventional models allows Aspen Plus to correctly determine the heating value and energy changes associated with pyrolysis and char combustion. A value for the heat of combustion was assumed to have been 20 MJ/kg following a class example.

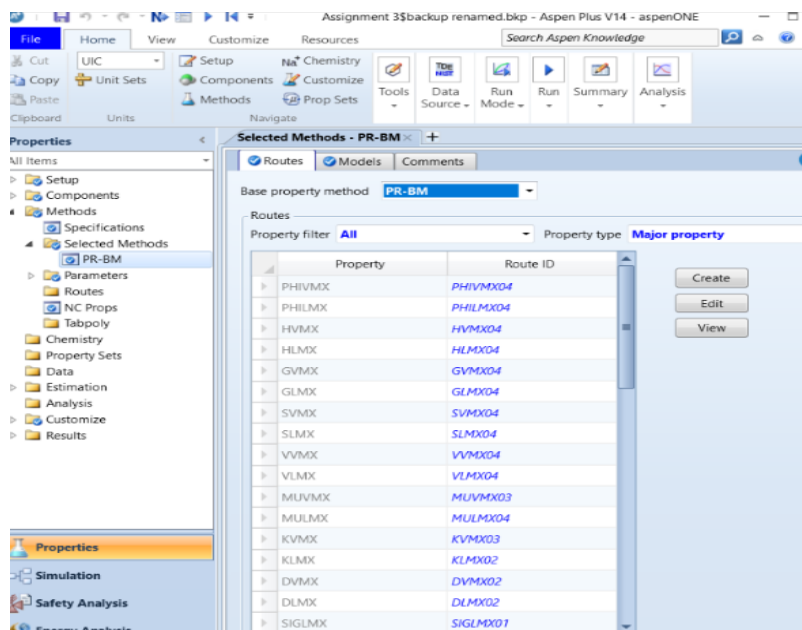


Figure 3 Physical Properties

Figure 3 displays the selection of the Peng Robinson with Boston Mathias (PR-BM). PR-BM accurately handles bio-oil vapors, permanent gases, and combustion mixtures, making it suitable for both the pyrolysis and flue gas streams.



Figure 4 PFD

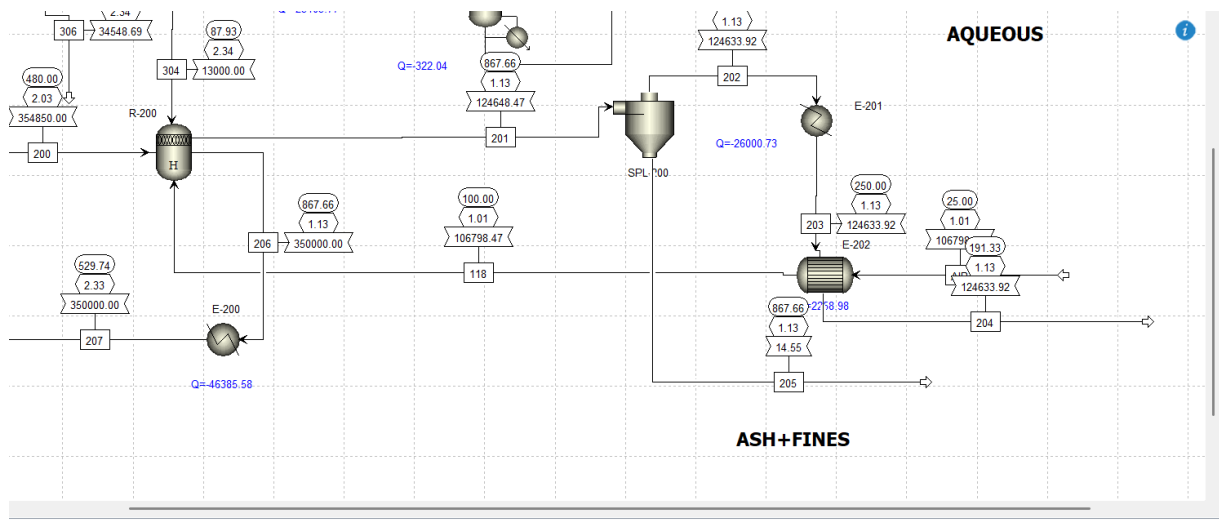


Figure 7 Reaction section

2-Heat and Weight Balance

The Heat and Weight balance for the pyrolysis reactor is as follows:

Assignment 3\$backup renamed - Copy.apw - Aspen Plus V14 - aspenONE

Batch Dynamics Plant Data Equation Oriented View Customize Reso

UIC Save Save as New Discard Change General Options Phases Stream Summary Options Flows Composition Properties

Capital: ___ USD Utilities: ___ USD/Year Energy Savings: ___ MW (___ %)

Main Flowsheet 100 (MATERIAL) - Results (Default)

Material	Vol. % Curves	Wt. % Curves	Petroleum	Polymers	Solids	Status
100 - Total Stream						
Temperature	C	25	375	529.738	480	
Pressure	bar	2.33048	2.34061	2.33048	2.0265	
Mass Flows	kg/hr	55555	35000	350000	440555	
Mass Fractions						
Enthalpy Flow	kW	-97118.4	-49590.2	-1.42361e+06	-1.56182e+06	
Energy Flow rate						
MIXED Substream						
Phase			Vapor Phase		Vapor Phase	
Temperature	C		375		480	
Pressure	bar		2.34061		2.0265	
Molar Vapor Fraction			1		1	
Mass Flows	kg/hr		35000		85705	
Mass Fractions						
Mole Flows	kmol/hr		0	914.089	0	2214.67
Mole Fractions						
Molar Enthalpy	kJ/kmol		-195304		-204316	
Molar Entropy	kJ/kmol-K		5.97009		-10.3138	
Enthalpy Flow	kW		-49590.2		-125658	
Average MW			38.2895		38.7092	
Molar Density	kmol/cum		0.0434685		0.0324049	
Heat capacity, mixture	kJ/kmol-K		64.8882		76.0156	
Availability flow, mixture	kW		-50042.2		-123767	
Energy Flow rate	kW		2374.92		10116.9	
SOLID Substream						
Temperature	C		25		480	
Pressure	bar		2.33048		2.0265	
Mass Flows	kg/hr		55555	0	0	4850
Mass Fractions						
Enthalpy Flow	kW		-97118.4			-6715.98

Figure 8 Heat and Weight Balance for Pyro reactor

Figure 8 demonstrates the heat and weight balance for the char combustor where streams 100,101,102 are the inlet streams and stream 104 is the outlet stream.

The heat and weight balance for the char combustor is as follows:

UIC

Save Save as New

Discard Change

Template

General Options

Stream Summary Options

Flows

Composition

Property Sets

Capital: ____USD Utilities: ____USD/Year

Energy Savings: ____MW (____%)

Exchangers -

Main Flowsheet

200 (MATERIAL) - Results (Default)

Material

Vol.% Curves

Wt. % Curves

Petroleum

Polymers

Solids

Status

</

Figure 9 Heat and Weight for Char Combustor

Figure 9 demonstrates the heat and weight balance for the char combustor where streams 200,304,207,208 are the input streams and streams 201 and 206 are the output streams. Only the mixed sub stream is shown for sake of simplicity.

The heat and weight balance for the separator SEP2 is as follows:

UIC

Save

Save as New

Discard Change

Template

 General Options

☒ Total

☒ Substreams

☐ Phases

Stream Summary Options

☒ Mole

☒ Mass

☐ Volume

Flows

☒ Mole

☒ Mass

Composition

 Select Properties

 Display Options

Property Sets

Capital: ___USD Utilities: ___USD/Year

Energy Savings: ___MW (___%)

Main Flowsheet x 300A (MATERIAL) - Results (Default) x +

Material

Vol.% Curves

Wt. % Curves

Petroleum

Polymers

Solids

 Status

	Units	300	300A	301	
- Total Stream					
Temperature	C	45	45	45	
Pressure	bar	1.72253	1.72253	1.72253	
+ Mass Flows	kg/hr	85705	47548.7	38156.3	
+ Mass Fractions					
Enthalpy Flow	kW	-155294	-74510.2	-81106.2	
Exergy flow rate					
- MIXED Substream					
Phase				Liquid Phase	
Temperature	C	45	45	45	
Pressure	bar	1.72253	1.72253	1.72253	
Molar Vapor Fraction		0.509741	0.909736	0	
+ Mass Flows	kg/hr	85705	47548.7	38156.3	
+ Mass Fractions					
+ Mole Flows	kmol/hr	2214.07	1263.39	950.683	
+ Mole Fractions					
Molar Enthalpy	kJ/kmol	-252503	-212315	-307129	
Molar Entropy	kJ/kmol-K	-119.2	-42.3228	-225.103	
Enthalpy Flow	kW	-155294	-74510.2	-81106.2	
Average MW		38.7092	37.6358	40.1357	
Molar Density	kmol/cum	0.128927	0.0725149	23.4357	
Heat capacity, mixture	kJ/kmol-K	74.6977	51.2307	105.666	
Availability (flow), mixture	kW	-133437	-70081.8	-63382.6	
Exergy flow rate	kW	446.963	432.845	18.6001	
+ CISOLID Substream					

Figure 10 Heat and Weight Balance for Quench Absorber

Figure 10 illustrates the heat and weight balance for the quench absorber where stream 300 is the inlet stream while stream 300A and 301 are the outlet streams. Once again only the mixed sub stream is shown for sake of simplicity. The heat and weight balance for the decanter is as follows:

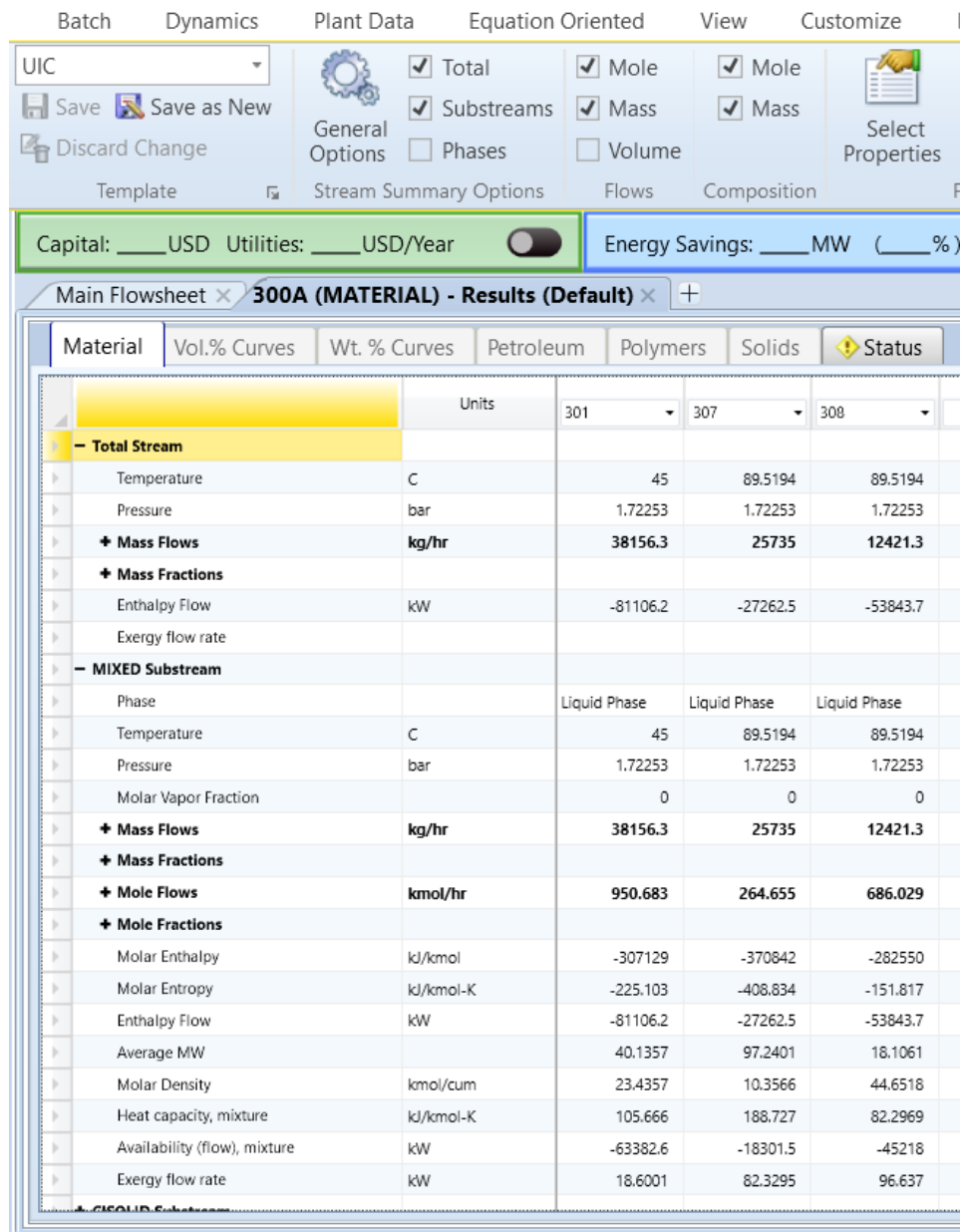


Figure 11 Heat and Weight balance for decanter

Figure 11 depicts the heat and weight balance for the decanter where stream 301 is the inlet and streams 307 is the bio-oil and 308 is the aqueous product.

3-Solution

Preparation

Before the biomass can be reacted it must be dried and prepared, and this is done within the hierarchy B1 as pictured below:

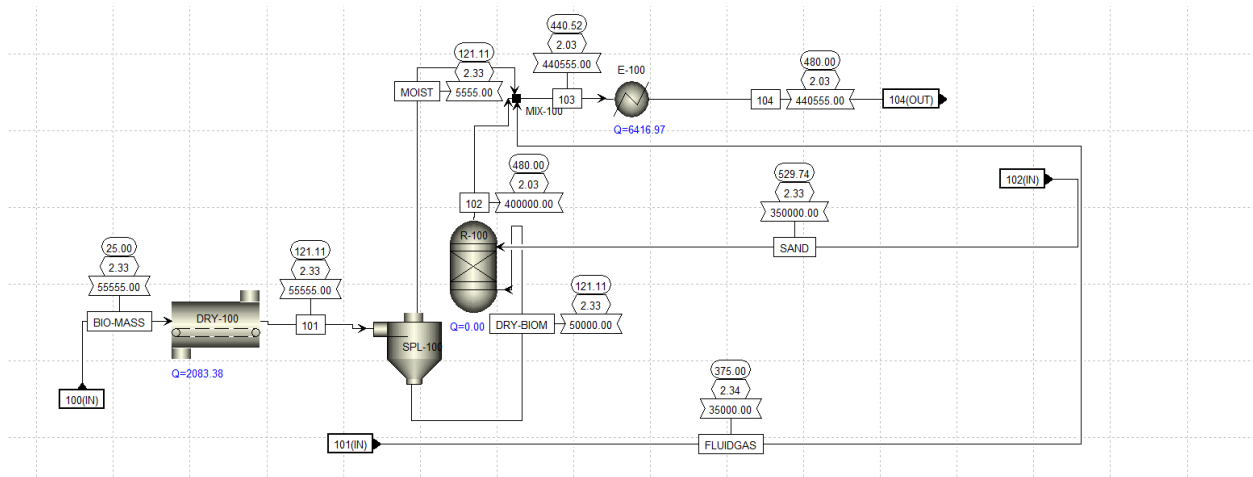


Figure 10 12 Feed Preparation

The amount of biomass specified in the problem statement is 50,000 kg/hr with a moisture content of 10%. To find the wet biomass incoming into the dryer excel goal seek was used to find the mass flow rate of water + biomass that would give a 10 wt% mixture. This amount 55,555 kg/hr was then input into a dryer and dried down into a very low moisture amount (.0001). Attempting to completely dry the solution resulted in errors within the dryer and unreasonably high duty values. The biomass is then split into the dry biomass which is sent into the reactor and then the moisture is which is sent into a mixer above the reactor. Both the sand and fluidizing gas are input into the pyrolysis reactor but attempting to do this inside the flowsheet resulted in convergence errors and mixing them above the reactor is a work-around. Since the temperature outside of the reactor does not match the temperature outside of the mixer a heater must be placed to reheat it up to the real reactor effluent reactor temperature of 480°C.

Both the mixer and heater are not present in real operations and are simply done for the sake of converging the flowsheet.

Reactor section

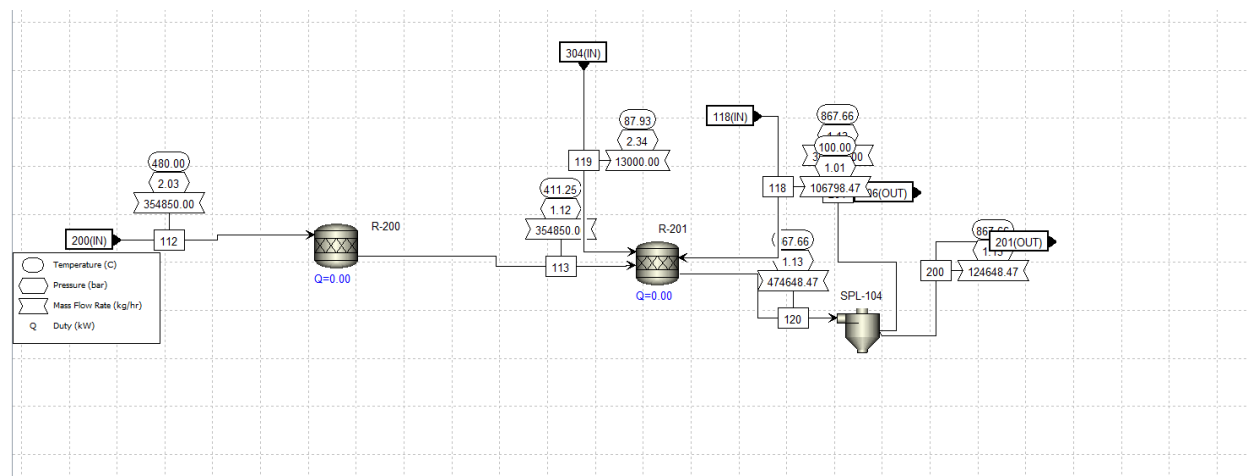


Figure 1113 Reaction Hierarchy

In Aspen Plus, the decomposition of biomass was modeled using an R-YIELD block. The model does not calculate reaction rates; it simply splits the biomass according to the experimental product yields given in the assignment. The pyrolysis reactor operates at 480°C and adiabatically produces bio-oil vapors, gases, water, and solid char. The biomass is decomposed into its components CO₂, H₂O, N₂, and SO₂ and char. The char produced in this step is going to be burned in a second R-STOIC reactor. To determine the stoichiometric coefficients for the components of char a calculator block was utilized. The molecular weights of the components of char were imported (C, H₂, N₂, O₂, Ash and S) and multiplied by their percentage inside the ultimate analysis provided in the problem statement. They were then divided by 100 to give the final coefficients and were exported into the first R-Stoic pictured in figure 11. The input to the second R-Stoic R-201 was the given operating pressure of 1.12 and adiabatic operation (duty=0). The final reactor R-201 was specified as a combustion reactor and no further specification is required. To reach the desired 2% O₂ composition in the combustor effluent a design spec was

utilized that modified the flow rate of air incoming to the char combustor to achieve the target of 2% O₂. This amount of O₂ is present because it is an excess amount of air which will help with combustion, and the heat that is coming from the char will be taken and sent back to the pyrolysis reactor through sand.

The sand absorbs heat in the combustor, and that heat gets released in the pyrolysis reactor. The temperature change of the sand streams accounts for heat transfer between the two units. The stream 207 and 102 are both the same sand stream but are not connected and are instead modeled as a semi-tear stream. A calculator block pushes the temperature of 207 into 102 and to ensure the reactor operated adiabatically another design spec was utilized that modified the temperature of the incoming sand stream to achieve zero duty. In reality, the streams are the same but attempting to connect them within the simulation leads to convergence errors, so this model was chosen instead. Following the reactor the sand is at a high temperature (867°C) and is cooled via steam generation. A cyclone (SPL-200) splits the gases from the solids where then the solids are sent out Ash and Fines while the gases are sent into a heat exchanger that cools them by heating incoming ambient air. The ambient air is then sent into the combustion reactor to reduce duty and provide an oxygen source. The gases are cooled to a reasonable temperature of 250°C via steam generation and are further cooled to 191°C via air cooling.

Separation Section

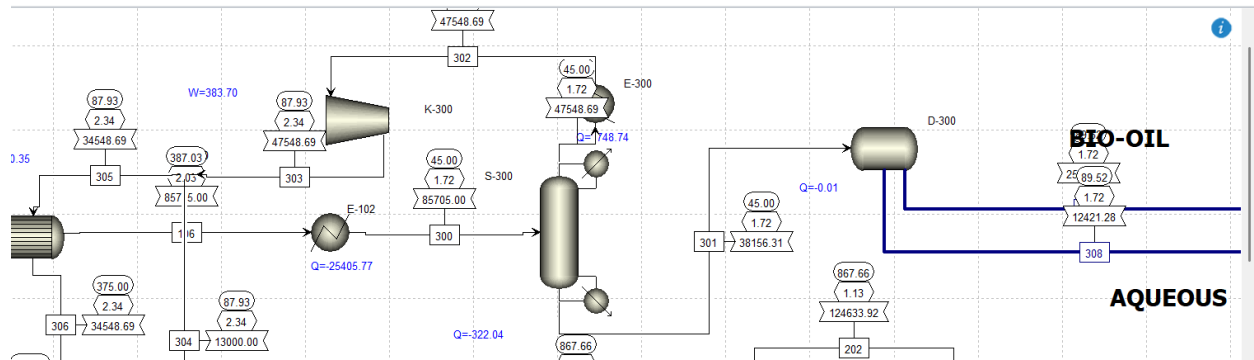


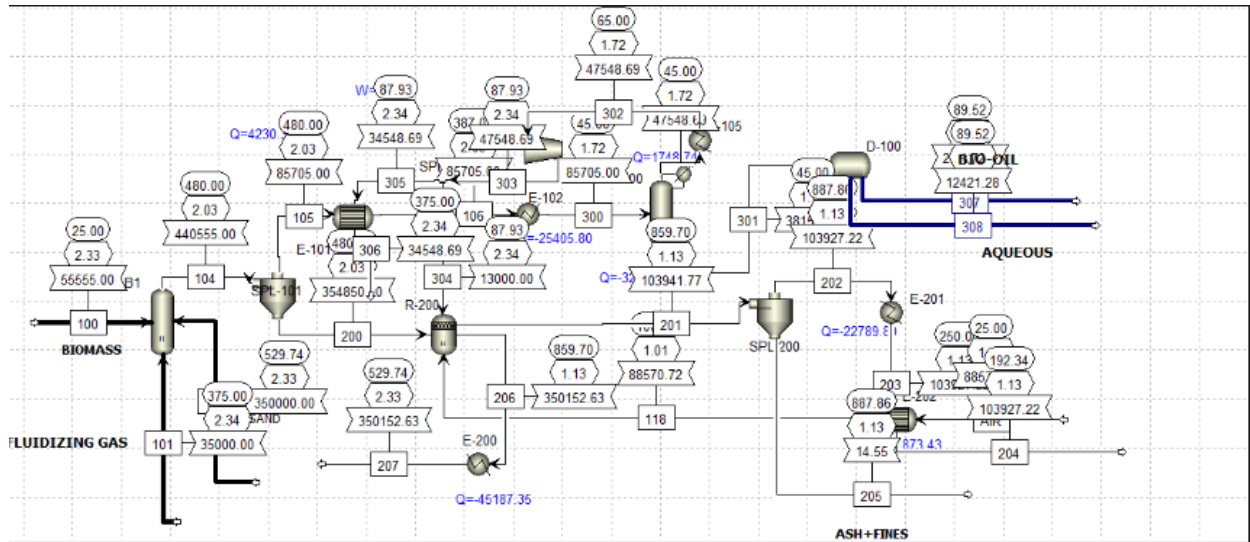
Figure 14 Separation Section

Following the heater E-102 the separation section of the flowsheet begins. Following the heat exchanger the stream is still at 387°C and must be further cooled to the given operating conditions of the SEP2 block of 1.7 atm and 45°C via steam generation from ambient water. Inside of SEP2 it is assumed that all the gases (H_2 , C_2 , N_2 , CO , CO_2 , NO_2 , SO_2 , CH_4 , C_2H_4 and .96 split of acetone and .2 split of water) rise out of the separator with the remaining bio-oil and aqueous solution fall out the bottom. The stream coming out of the top of the separator is fed into turbine K-300. Prior to that it must be set at its dew point to avoid errors in the software and damage the compressor. The heater is specified so that the stream exiting is 65°C and a pressure drop of 0 is assumed. The value should realistically be 1-5 psi but it depends on the type of heat exchanger which is not specified in the problem. The temperature of 65 was found through trial and error. Any lower temperature would not return to the solution at its dew point and any higher would simply increase the duty. For compressor K-300 a typical value for isentropic efficiency .8 was assumed and a mechanical efficiency of 1 was assumed due to lack of information. The outlet pressure of the compressor is set to the inlet pressure of the fluidizing gas of 2.31 atm as this is the origin of the fluidizing gas.

Following the turbine the stream is split into the fluidizing gas and the bleed gas to the combustor. The splitter is specified so that the given amount of bleed gas to the reactor 13,000 kg/hr is achieved and the remainder is sent through heat exchanger E-101. The fluidizing gas is the remainder and should be exactly 35,000 kg/hr as specified in the problem statement. This is where the stream should connect back to the pyrolysis reactor but attempting to connect them on the flowsheet causes convergence errors, so they are left as two-separate streams on the flowsheet streams 306 and 101.

The liquid product is taken from the bottom of the separator and sent to a decanter. The decanter splits the bio-oil product and the aqueous solution. D-300 is modeled as an ideal decanter with 0 pressure drop and 100% efficiency. To determine an actual yield for a real decanter the efficiency would have to be known alongside the pressure drop which are determined by the design of the vessel.

With a complete flowsheet the duties and yields can be calculated. The duties are taken exactly from Aspen and used to calculate the amount of steam generated using the heat of vaporization to determine the amount of steam produced from ambient water $T=25^{\circ}\text{C}$ and $P=1\text{ atm}$. The recoveries are calculated by determining the mass fraction of Carbon in each component and multiplying that by the mass flow rate of carbon in each stream to determine the amount of carbon in each stream. This is repeated for each desired recovery (Bio-Oil, Flue-Gas and Wastewater).



Results

Steam Generation Calculation

Heat recovered in the circulating pyrolysis system is used to produce steam at two pressure levels. In this process, hot sand coming from the char combustor transfers energy back into the system; while cooling and condensation of vapors release extra heat. Water is going to be coming at about 25°C and it will absorb this recovered heat and after which it will vaporize to form saturated steam at the temperatures of 250°C high pressure steam and 125°C of low-pressure steam. Aspen reports on the heat duties from the two exchangers that will supply heat for steam generation. The combustor cooler is going to recover most of the heat, providing 25405.8 Kw, which will be enough to heat and vaporize water into 250°C high pressure steam. Additional heat is going to be recovered upstream of the product separator where the condensation of pyrolysis vapors will provide 748.78 Kw, Producing 125°C low pressure steam. To determine how much steam is produced, the heat duties were divided by the enthalpy required to convert liquid water around 25°C into saturated steam at each temperature. Using T_{sat} or we can also use the steam data, the required energy changes

are going to be around 2880KJ/Kg for 250°C steam and 2620 KJ/KG for 125°C steam. $m =$

$$\frac{2880}{25405.8} = 8.8 \frac{\text{kg}}{\text{s}} m = \frac{2880}{25405.8} = 8.8 \frac{\text{kg}}{\text{s}}.$$

The mass flow rates were converted to volumetric flow rates using the ideal gas at standard conditions and that gave an answer of

33.5 MMSCFD of high-pressure steam, and 1.1 MMSCFD of low-pressure steam. This shows that steam production is mostly reliant on the combustor's side, and the condensation of pyrolysis vapors also helps with production but in smaller amounts of low-pressure steam.

The yields for the bio-oil are as follows:

Bio-oil	Flow	Carbon Frac	Carbon flow
Acetone	853.9517	0.6204025	529.79374
Acid	5503.828	0.4000163	2201.6206
Furfural	1443.8987	0.6250159	902.4596
m-cresol	1999.6178	0.7774835	1554.6698
Guaiacol	13739.701	0.6772794	9305.6161
Levogluco	2182.4521	0.4444612	970.01522
CTOT			15464.175
kg/hr			
CIN	24830		
kg/hr			
CYIELD	62%		

Figure 15 Carbon Yield for bio-oil

Figure 15 depicts the carbon yield for bio-oil which is 62%.

The same process described in the solution section is repeated for the bio-oil, flue-gas and wastewater. The yield for the flue gas is as follows:

YIELD FLUE GAS	FLOW	CarbonFrac	Carbon Flow
CO2	30284.655	0.2729165	8265.1818
Cyield	27%		

Figure 16 Flue gas Yield

Figure 16 depicts the yield for flue gas which is 27% carbon yield. The yield for wastewater is as follows:

Wastewater	Flow	Carbon Frac	Carbon flow
Acetone	0.3939509	0.6204025	0.2444081
Acid	46.173395	0.4000163	18.470109
Furfural	6.1016711	0.6250159	3.8136413
m-cresol	0.3827201	0.7774835	0.2975586
Guaiacol	10.302384	0.6772794	6.9775919
Levogluco	17.548422	0.4444612	7.7995919
CTOT	80.902543		37.6029

Figure 17 Yield for Wastewater

Figure 17 depicts the yield for the wastewater, which is .15% which is ideal. If it were high, it would indicate that we are losing product in the waste stream which is undesirable.

Stream	Yield
BIO-OIL	62%
FLUE GAS	27%
Wastewater	0.1514%

Figure 18 Final yields

The final yields are presented in a table in figure 18.

The only unit operation taking power is the compressor which has a calculated duty of 383 KW.

WTOT	383
KW	

Figure 19 Calculated Duty

References

Aspen Technology, Inc. *Aspen Plus, Version 14*; AspenTech: Bedford, MA, 2004.

Caaracostis, D. (2025). Assignment #3 (CHE 433 course) document

